An Energy-Based View on the Manifold Hypothesis

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Abstract

A commonly adopted assumption in modern machine learning is the *manifold hypothesis*, which claims that although data is embedded in a high-dimensional ambient space, the probability mass concentrates near a much lower-dimensional geometric structure. However, the real-world latent variables that images are drawn from are continuous, not discrete, and therefore cannot be completely captured by a finite manifold. Moreover, the assumption does not properly capture the union of manifolds, stratified spaces, and manifolds with singularities and boundaries. Drawing upon recent advances in energy-based models, we provide an energy-based perspective on the real-world latents, describing an Occam's razor style analog for the manifold hypothesis, which is more amenable to recent literature in the field. We theoretically prove that our construction recovers the manifold hypothesis for coarse functions, and validate it with numerical examples.

1 Introduction

If pictures are discretized objects in $\mathbb{R}^d$, where $d = n^2$ is the dimension of the image.

We consider raw objects $f : [0, 1]^2 \to \mathbb{R}^3$ corresponding to depixelated RGB pictures.

A commonly adopted assumption in modern machine learning is the *manifold hypothesis*, which claims that although data is embedded in a high-dimensional ambient space, the probability mass concentrates near a much lower-dimensional geometric structure.

Assumption 1.1 (Manifold Hypothesis)**.** Let $\chi \subset \mathbb{R}^D$ be the ambient space and let $\mathcal{D}$ be a probability distribution on χ. There exists a d-dimensional manifold $\mathcal{M} \subset \chi$, with $d \ll D$, such that

$$\mathbb{P}_{x\sim\mathcal{D}}\big(\mathrm{dist}(x, \mathcal{M}) \leq \varepsilon\big) \geq 1 - \delta,$$

for sufficiently small $\varepsilon, \delta > 0$.

In practice, data rarely concentrate on a single smooth manifold. Rather, they are better modeled as lying on *unions of manifolds*, or more generally on *stratified spaces* composed of pieces of varying intrinsic dimensions, potentially with singularities and boundaries.

Why small d is important?

- Generalization error depends on the intrinsic dimension d of the manifold.
- The number of samples n needed to learn the manifold increases with d.
- The complexity of the model increases with d.
- Can get a small embedding.

Some shortcomings of the manifold hypothesis are that

- Doesn't properly capture union of manifolds.

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- Doesn't properly capture stratified spaces.
- Doesn't properly capture manifolds with singularities and boundaries.
- What about infinite-dimensional data? Cause images come from infinite-dimensional continuous functions.

[Emile says: would be nice to elaborate on the above, and to point to specific instances where assuming it has some "catastrophic" consequences. that could go into a motivating examples section]

1.1 Related Works

Energy-based models Song and Kingma [2021], Du and Mordatch [2019], LeCun et al. [2006], Zhu et al. [2024], Du et al. [2021, 2020]

Compositional generation Su et al. [2024], Liu et al. [2021, 2022, 2023], Du and Kaelbling [2024], Bagautdinov et al. [2018]

Occam's razor Elmoznino et al. [2025], Rasmussen and Ghahramani [2000]

Example: Vision Data Each 2d image (or 3d scene or a point cloud) can be modeled $f : \mathbb{R}^2 \to \mathbb{R}^3$ as a function from coordinates to RGB (or XYZ) values. That's how NeRFS work:

Neural Radiance Fields (NeRFs). Mildenhall et al. [2021], Pumarola et al. [2021], Xu et al. [2021], Wang et al. [2022], Rabby and Zhang [2024], Reiser et al. [2021], Müller et al. [2022], Tancik et al. [2021], Yu et al. [2021], Chen et al. [2022] [Emile says: this actually looks very relevant. would be worth doing a deeper dive into this] A *Neural Radiance Field* models a 3D scene as a continuous function

$$f_\theta : (\mathbf{x}, \mathbf{d}) \;\mapsto\; (\sigma, \mathbf{c}),$$

where

- $\mathbf{x} \in \mathbb{R}^3$ denotes a spatial location,
- $\mathbf{d} \in \mathbb{S}^2$ denotes a viewing direction,
- $\sigma \in \mathbb{R}_{\geq 0}$ is the volume density at $\mathbf{x}$,
- $\mathbf{c} \in \mathbb{R}^3$ is the emitted RGB color in direction $\mathbf{d}$.

The key idea is that the scene is represented neither as a mesh, nor as a voxel grid, nor as a point cloud, but as a *continuous radiance field* parameterized by a neural network.

In a similar way, each 2d image can be modeled as a function from coordinates to RGB values: $f : \mathbb{R}^2 \to \mathbb{R}^3$ and the embedding of this function would the weights of the corresponding ReLU network. The regular manifold hypothesis doesn't properly capture this structure that:

- In a union of manifolds
- Has some hierarchical structure.
- Has permutation-invariance with respect to weights in each layer.

Also, importantly, each image f is not a smooth function: its a combination of a "simple" in some sense partition of the image into smooth enough regions.

2 Problem Setting

Notation. We refer to a metric space (X, d) as a Polish space if it is complete (every Cauchy sequence in (X, d) converges to a point in X) and separable (there exists a countable dense subset $D \subset X$). For instance, $\mathbb{R}^n$ with the Euclidean metric, separable Banach spaces, Hilbert spaces, and function spaces such as $L^2(M)$ or $H^s(M)$ (when M is a compact manifold) are all Polish spaces. Moreover, if X is Polish, then $(X, \mathcal{B}(X))$ enjoys strong regularity properties that make it a canonical measurable structure in probability theory, where $\mathcal{B}(X)$ is the Borel σ-algebra on X.

Supervised Learning. From data $D = \{(x_i, y_i)\}_{i=1}^n$ we seek to learn a target function $f^* : \mathcal{X} \to \mathbb{R}$ such that $f^*(x_i) = y_i$ for all $i = 1, \dots, n$. Usually, we know the input space $\mathcal{X}$ via some base distribution ν_0 on $\mathcal{X}$, for instance, uniformly distribution over a compact set $\mathcal{X} \subset \mathbb{R}^d$. Moreover, we assume that smoother or more predictable the function f^* is, the better it generalizes to new data.

Definition 2.1 (Symmetric Dirichlet Space)**.** Let $(X, \mathcal{F}, \mu)$ be a σ-finite measure space. A *Dirichlet form* on $L^2(\mu)$ is a pair $(\mathcal{E}, \mathcal{D}(\mathcal{E}))$ such that:

1. $\mathcal{D}(\mathcal{E}) \subset L^2(\mu)$ is a dense linear subspace;
2. $\mathcal{E} : \mathcal{D}(\mathcal{E}) \times \mathcal{D}(\mathcal{E}) \to \mathbb{R}$ is a symmetric bilinear form;
3. (Non-negativity)
$$\mathcal{E}(f, f) \geq 0 \qquad \forall f \in \mathcal{D}(\mathcal{E});$$
4. (Closedness) $\mathcal{D}(\mathcal{E})$ is complete with respect to the norm
$$\|f\|_{\mathcal{E}}^2 := \|f\|_{L^2(\mu)}^2 + \mathcal{E}(f, f);$$
5. (Markov property) for every $f \in \mathcal{D}(\mathcal{E})$, the truncation $\widetilde{f} := (0 \vee f) \wedge 1$ satisfies
$$\widetilde{f} \in \mathcal{D}(\mathcal{E}) \quad \text{and} \quad \mathcal{E}(\widetilde{f}, \widetilde{f}) \leq \mathcal{E}(f, f).$$

The triple $(X, \mathcal{F}, \mu, \mathcal{E})$ is called a *Symmetric Dirichlet Space*.

The Dirichlet form encodes the geometry of the learning space via the energy functional: $\mathcal{E}(f, f)$ is the energy of the function f. Enforcing the Markov property allows us to define stochastic processes on the learning space: semi-group generated by $\mathcal{E}$ is a Markov semigroup.

The main idea of learning is that if $x_1 \approx x_2$ then $f(x_1) \approx f(x_2)$ for a function f. Dirichlet form enforces precisely this property. As it encodes the geometry of the learning space, it allows us to define a generalized Laplacian L on $L^2(\mu)$ and Sobolev classes of functions on X.

Definition 2.2 (Generalized Laplacian)**.** Let $(X, \mathcal{F}, \mu, \mathcal{E})$ be a measure space endowed with a Dirichlet form. There exists a unique self-adjoint operator L on $L^2(\mu)$ such that
$$\mathcal{E}(f, g) = \langle f, Lg \rangle_{L^2(\mu)}, \qquad \forall\, f \in \mathcal{D}(\mathcal{E}),\ g \in \mathcal{D}(L).$$
The operator L admits a spectral decomposition $L\phi_j = \lambda_j \phi_j$, with
$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots, \qquad \langle \phi_j, \phi_k \rangle_{L^2(\mu)} = \delta_{jk}.$$

The Markov property of the Dirichlet form $\mathcal{E}$ implies that e^{tL} is a Markov semigroup. Hence, our distribution μ is a stable state of the system: given a base distribution ν_0 on $\mathcal{X}$, the distribution μ is the limit of the Markov process $e^{tL}\nu_0$ as $t \to \infty$.

Definition 2.3 (Effective dimension)**.** The *effective dimension* of the learning space $(X, \mathcal{F}, \mu, \mathcal{E})$ is the quantity $d_{\text{eff}} \in (0, \infty)$ such that the eigenvalues of the corresponding Laplacian operator $L : L^2(\mu) \to L^2(\mu)$ satisfy
$$\lambda_j \asymp j^{2/d_{\text{eff}}} \qquad \text{as } j \to \infty,$$
or equivalently, the eigenvalue counting function satisfies
$$N(\lambda) := \#\{j : \lambda_j \leq \lambda\} \asymp \lambda^{d_{\text{eff}}/2} \qquad \text{as } \lambda \to \infty.$$

Or, more generally
$$\operatorname{tr} e^{tL} \sim t^{-d_{\text{eff}}/2} \quad \text{as } t \to 0^+$$

3 Set Up

Common belief about the data but make it abstract:

- Observed data $x_1, \ldots, x_n$ comes from some space X (e.g. $\mathbb{R}^D$)
- We know some "sparse" distribution over X: ν_0 (e.g. Gaussian, Uniform)
- However, we believe that the data comes from some "compact" or "concentrated" distribution μ over X (e.g. d-dim manifold $M \subset \mathbb{R}^D$)

If that is the case, we have better hope to learn functions $f : X \to \mathbb{R}$: more concentration – faster rates.

First, define the data space and equip it with the differentiation property.

- Let X be a set
- an algebra of functions $f : X \to \mathbb{R}$, $\mathcal{A}$
- $D : \mathcal{A} \to \mathcal{M}$ a derivation into some module $\mathcal{M}$ equipped with a pointwise inner product $\langle \cdot, \cdot \rangle_x$
- Leibniz rule: $D(fg) = fDg + gDf$.
- Hence, the carre du champ operator is $\Gamma(f, g)(x) = \langle Df, Dg \rangle_x$. It generalizes square-gradient concept.

Now let us add a measure ν over X. That defines:

- Dirichle form $\mathcal{E}_\nu(f, g) = \int_X \Gamma(f, g) d\nu(x)$ ([TODO: maybe add some assumptions on this])
- Self-adjoint Laplacian L_ν such that $\mathcal{E}_\nu(f, f) = \langle f, L_\nu f \rangle_\nu$
- Markov semi-group: e^{-tL_ν}, it defines a diffusion that converges to ν.
- The iterated carré du champ (or Γ_2^ν operator) is:

$$\Gamma_2^\nu(f, g) = \frac{1}{2}(L_\nu \Gamma(f, g) - \Gamma(f, L_\nu g) - \Gamma(g, L_\nu f))$$

  and

$$\Gamma_2^\nu(f) := \Gamma_2^\nu(f, f) = \frac{1}{2}\left(L_\nu |Df|^2 - 2\langle Df, D(L_\nu f) \rangle\right)$$

Alright, what defines the "concentration" of a measure ν? What structural assumption would give us hope to learn functions on it?

4 Key Assumptions

Definition 4.1 (Curvature–Dimension Condition)**.** For $\rho \in \mathbb{R}, N \in (0, \infty)$, ν satisfies $CD(\rho, \Gamma)$ if for all $f \in \mathcal{A}$

$$\Gamma_2^\nu(f, f) \geq \rho \Gamma(f, f) + \frac{1}{N}(L_\nu f)^2$$

Intuitively: $\rho \sim$ curvature parameter, $N \sim$ dimension

4.1 Sub-Exponential Concentration

Definition 4.2 (Log-Sobolev Inequality)**.** Given μ for there is a constant C_{LS} such that the following holds for any $f : X \to \mathbb{R}$

$$\operatorname{Ent}_\mu\left(f^2\right) \leq 2C_{LS} \int \|Df\|^2 d\mu$$

where $\operatorname{Ent}_\mu(g) = \int g \log g d\mu - \int g d\mu \cdot \log \int g d\mu$ for $g \geq 0$.

Lemma 4.3. *Assuming* $\Gamma_2^\mu(f, f) \geq \rho\Gamma(f, f)$ *for all* $f \in \mathcal{A}$ *then* $C_{LS} \leq \frac{1}{\rho}$

Consequences:

Theorem 4.4 (Hypercontractivity, Gross 1975)**.** *The LSI is equivalent to hypercontractivity of the semigroup: for* $1 < p < q < \infty$*,*

$$\|P_t f\|_{L^q(\mu)} \leq \|f\|_{L^p(\mu)} \quad \textit{when } t \geq \frac{C_{LS}}{2} \log \frac{q-1}{p-1}$$

Theorem 4.5 (Sub-Gaussian concentration)**.** *For* f *with* $\Gamma(f,f) \leq 1$*,*μ*–a.e., the Herbst argument gives*

$$\mu\left(f - \mathbb{E}_\mu f \geq t\right) \leq \exp\left(-\frac{t^2}{2C_{LS}}\right)$$

The smaller C_{LS} – the better.

4.2 Effective Dimension

Definition 4.6 (Effective dimension)**.** The *effective dimension* of $(X, \mathcal{A}, \Gamma, \mu)$ is the quantity $d_{\text{eff}} \in (0, \infty)$ such that the eigenvalues of the corresponding Laplacian operator $L : L^2(\mu) \to L^2(\mu)$ satisfy

$$\lambda_j \asymp j^{2/d_{\text{eff}}} \qquad \text{as } j \to \infty,$$

or equivalently, the eigenvalue counting function satisfies

$$N(\lambda) := \#\{j : \lambda_j \leq \lambda\} \asymp \lambda^{d_{\text{eff}}/2} \qquad \text{as } \lambda \to \infty.$$

Or, more generally

$$\operatorname{tr} e^{tL} \sim t^{-d_{\text{eff}}/2} \quad \text{as } t \to 0^+$$

The smaller d_{eff} – the better.

Corollary 4.7. *If* ν *satisfies* $CD(\rho, \Gamma)$ *then* $d_{\text{eff}} \leq N$

5 Supervised Learning: Learning a Function From Data

Adding a prior $p(f) \propto e^{-E(f)}$ over functions $f : X \to \mathbb{R}$ the supervised learning problem is equivalent to the following optimization problem:

$$\widehat{f}_n = \min_f \sum_{i=1}^n \|f(x_i) - y_i\|^2 + E(f)$$

Note, that $E(f)$ defines sublevel sets $\mathcal{F}_r = \{f : E(f) \leq r\}$. The generalization error of $\widehat{f}_n$ decomposes as:

$$\mathbb{E}\left\|f^* - \widehat{f}_n\right\|_{L^2}^2 \leq \underbrace{\inf_{f : E(f) \leq r} \|f - f^*\|^2}_{\text{bias: how well } \mathcal{F}_r \text{ approximates } f^*} + \underbrace{\varepsilon_n^2(r)}_{\text{variance: estimation error within } \mathcal{F}_r}$$

Definition 5.1 (Sublevel set)**.** Given a convex functional $E : \mathcal{A} \to \mathbb{R} \cup \{+\infty\}$, a radius $r > 0$, and a metric d on $\mathcal{A}$, the sublevel set is:

$$\mathcal{B}_r(E) = \{f \in \mathcal{A} : E(f) \leq r\}$$

Definition 5.2 (Covering number)**.** The ϵ-covering number is:

$$N\left(\epsilon, \mathcal{B}_r(E), d\right) = \min\left\{m \in \mathbb{N} : \exists f_1, \dots, f_m \in \mathcal{A} \text{ s.t. } \mathcal{B}_r(E) \subseteq \bigcup_{i=1}^m \{g : d\left(g, f_i\right) \leq \epsilon\}\right\}$$

The natural metric in our setting is $d(f,g) = \|f - g\|_{L^2(\mu)} = \left(\int_X (f-g)^2 d\mu\right)^{1/2}$

Definition 5.3 (Metric entropy)**.**

$$\mathcal{H}\left(\epsilon, \mathcal{B}_r(E), d\right) = \log N\left(\epsilon, \mathcal{B}_r(E), d\right)$$

The smaller the metric entropy – the less samples we need to learn a function from data.

Lets consider 3 simple enough cases of what $E(f)$ can be:

1. Given convex $g : \mathbb{R} \to \mathbb{R}$ define $E(f) = \int_X g(f(x))d\mu(x)$
2. $E(f) = \mathcal{E}_\mu(f, f)$
3. Given $\phi : \mathbb{R} \to \mathbb{R}$ define $E(f) = \langle f, \phi(L_\mu)f\rangle$

Proposition 5.4. *For $E(f) = \int_X g(f(x))d\mu(x)$ The covering number of $\mathcal{B}_r(E)$ is at most $N(r, E) = \frac{1}{r}\int_0^r \mathcal{H}(\epsilon, \mathcal{B}_r(E), d)d\epsilon$.*

As we introduced Definition 4.1 that captures the "compactness" of the measure, we can say now that $\mu \ll \nu$ if

$$\mu \in CD(\rho_\mu, N_\mu), \nu \in CD(\rho_\nu, N_\nu)$$
and
$$\rho_\mu > \rho_\nu, N_\mu < N_\nu$$

The following theorem captures what density $\frac{d\mu}{d\nu}$ "simplifies" the measure.

Theorem 5.5 (Perturbation of $\mathrm{CD}(\rho, N)$)**.** *[TODO: double check, add a proof, better notation]*

Let $(X, \mathcal{A}, \Gamma, \mu_0)$ be a Markov diffusion structure with generator L_0 satisfying $\mathrm{CD}(\rho, N)$:

$$\Gamma_2^{(0)}(f, f) \geq \rho\Gamma(f, f) + \frac{1}{N}\left(L_0 f\right)^2 \quad \textit{for all } f \in \mathcal{A}$$

Let $U \in \mathcal{A}$ satisfy:

- *Abstract κ-convexity:* $\quad \Gamma(f, \Gamma(U, f)) - \frac{1}{2}\Gamma(U, \Gamma(f, f)) \geq \kappa\Gamma(f, f) \quad$ *for all f*
- *Bounded abstract gradient:* $\quad \Gamma(U, U) \leq M \quad \mu_0$*-a.e.*

Let $d\mu = Z^{-1}e^{-U}d\mu_0$ with $L_U f = L_0 f - \Gamma(U, f)$. Then for any $\varepsilon \in (0, 1)$, (L_U, Γ, μ) satisfies $\mathrm{CD}\left(\rho', N'\right)$ with

$$\rho' = \rho + \kappa - \frac{(1/\varepsilon - 1)M}{N}, \quad N' = \frac{N}{1 - \varepsilon}$$

In particular, $C_{LS}(\mu) \leq \frac{1}{\rho'}$ whenever $\rho' > 0$.

6 Diffusion Models

[TODO: Add noise schedule maybe.]

Forward process: $\mu \to \mu_0$ Density $p_t = \mu_t/\mu_0$ evolves according to the diffusion equation:

$$\frac{\partial p_t}{\partial t} = L_{\mu_0}p_t \implies p_t = e^{tL_{\mu_0}}\mu_0$$

Backward process: $\mu_0 \to \mu$ Density p_t evolves according to the diffusion+drift equation:

$$\frac{\partial p_t}{\partial t} = L_{\mu_0}p_t - \nabla \cdot (p_t \nabla \log p_t)$$

To be able to reverse it back, we need to learn $\nabla \log p_t$. Let us look at the very first δ step of the forward process (corresponding to the very last δ step of the backward process) and learn a score function for that step $s(\cdot) = \nabla \log p_T = \nabla \log \frac{d\mu}{d\nu}$. The minimization problem is:

$$\min_s \underbrace{\mathbb{E}_{p_T}\left[\|s(\cdot) - \nabla \log p_T\|^2\right]}_{\text{data term}} + \underbrace{R(s)}_{\text{regularizer term}} \tag{1}$$

While the data term is straightforward, the regularizer term is to be chosen. Assuming $d\mu = Z^{-1}e^{-U}d\mu_0 \Rightarrow \nabla \log p_T = -DU$. Our prior (hope) for U is that it concentrates the measure better, specifically, that by Definition 4.1 $U \in \mathcal{A}$ satisfies:

- κ-convexity: $\quad \Gamma(f, \Gamma(U, f)) - \frac{1}{2}\Gamma(U, \Gamma(f, f)) \geq \kappa\Gamma(f, f) \quad$ for all f
- Bounded gradient: $\quad \Gamma(U, U) \leq M \quad \mu_0$-a.e.

Define the abstract Hessian as a map from 1-forms to bilinear forms on $D(\mathcal{A})$:

$$H_v(f) := \langle Df, D\langle v, Df\rangle\rangle - \frac{1}{2}\langle v, D\langle Df, Df\rangle\rangle$$

This is linear in v : $H_{\alpha v_1 + \beta v_2}(f) = \alpha H_{v_1}(f) + \beta H_{v_2}(f)$. Note, that the term $\langle v, v\rangle \leq M$ is quadratic in v.

Hence, the following set is convex:

$$\mathcal{C}_{\kappa,M} = \{v \in D(\mathcal{A}) : H_v(f) \geq \kappa\Gamma(f, f) \text{ and } \langle v, v\rangle \leq M \quad \forall f \in \mathcal{A}\}$$

First let us compute its covering number.

Lemma 6.1. *The covering number of $\mathcal{C}_{\kappa,M}$ is at most $N(M, \kappa) = \frac{M}{\kappa}$.*

Lemma 6.2 (Very rough sketch)**.** *Define $W_{\kappa,M}$ as [TODO: a fixed set of very cool points from $\mathcal{C}_{\kappa,M}$]. Then for every $v \in \mathcal{C}_{\kappa,M}$ the following problem has a unique solution:*

$$\min_{\omega \in \mathfrak{M}_+(W_{\kappa,M})} \|\omega\|_{TV} \text{ such that } \int_{w \in W_{\kappa,M}} w d\omega(w) = f$$

Hence, if we know the set $W_{\kappa,M}$ then Equation (1) can be rewritten as:

$$\min_{\omega \in \mathfrak{M}_+(W)} \mathbb{E}_{p_T}\left[\left\|\int_{w \in W_{\kappa,M}} w d\omega(w) - \nabla \log p_T\right\|^2\right] + \|\omega\|_{TV}$$

6.1 References

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7 Energy-Based Model Perspective on Effective Dimension

need framework that shows that

- pure noise

- fractals

- matryoshka

should be hard to learn

7.1 ASTs

Definition 7.1 (AND–OR Abstract Syntax Tree)**.** Let $\mathcal{X}$ be an input space and $\phi : \mathcal{X} \to \mathbb{R}^d$ an embedding. An *AND–OR abstract syntax tree (AST)* is defined recursively as:

- **Atom (Leaf):**
$$T(x) = g(x), \qquad g \in \mathcal{G},$$
where $\mathcal{G}$ is a class of simple local models (e.g. linear or affine functions).
- **AND node:**
$$T(x) = \mathbb{I}\big[C_1(x) \wedge \cdots \wedge C_k(x)\big] \cdot T'(x),$$
where each $C_i(x)$ is a constraint and T' is a subtree.
- **OR node:**
$$T(x) = \sum_{j=1}^{m} \mathbb{I}\big[C_j(x)\big] \cdot T_j(x),$$
where $\{C_j\}_{j=1}^m$ form a partition of $\mathcal{X}$ and T_j are subtrees.

The constraints $C(x)$ are typically hyperplane tests of the form
$$C(x) := \langle w, \phi(x) \rangle \geq 0.$$

Semantics.

- AND nodes represent conjunctions of constraints and correspond to intersections of half-spaces or cross-products of independent factors.
- OR nodes represent partitions of the input space into disjoint regions.
- Each root-to-leaf path defines a region
$$\mathcal{X}_\ell = \bigcap_{C \in \text{path}(\ell)} C,$$
on which the function reduces to a simple local model.

Example: 1D Piecewise Linear Function Let
$$f(x) = \begin{cases} a_1 x + b_1, & x \leq t, \\ a_2 x + b_2, & x > t. \end{cases}$$
This function admits the AND–OR representation
$$f(x) = \mathbb{I}[x \leq t]\,(a_1 x + b_1) + \mathbb{I}[x > t]\,(a_2 x + b_2).$$
Equivalently, as an AST:
$$\texttt{OR}\Big(\texttt{AND}(x \leq t,\ \texttt{LINEAR}(a_1, b_1)), \texttt{AND}(x > t,\ \texttt{LINEAR}(a_2, b_2))\Big).$$

Example: Hierarchical Partition Let $t_1 < t_2$ and
$$f(x) = \begin{cases} a_1 x + b_1, & x \leq t_1, \\ a_2 x + b_2, & t_1 < x \leq t_2, \\ a_3 x + b_3, & x > t_2. \end{cases}$$
Then f can be written as
$$f(x) = \mathbb{I}[x \leq t_2]\Big(\mathbb{I}[x \leq t_1](a_1 x + b_1) + \mathbb{I}[t_1 < x \leq t_2](a_2 x + b_2)\Big) + \mathbb{I}[x > t_2](a_3 x + b_3).$$
This corresponds to a depth–2 AND–OR tree.

Example: ReLU as AND–OR For $w \in \mathbb{R}^d$,

$$\mathrm{ReLU}(\langle w, x\rangle) = \mathbb{I}[\langle w, x\rangle \geq 0]\langle w, x\rangle + \mathbb{I}[\langle w, x\rangle < 0] \cdot 0.$$

Thus a ReLU unit is an OR node with two AND branches. A finite ReLU network corresponds to a (possibly large) AND–OR tree, whose leaves are linear functions and whose OR structure encodes the activation pattern partition.

[Emile says: What kind of non-trivial theoretical results to validate this construction would be the most insightful? One thing is that for coarse/non-granular functions, it should sort of recover the original manifold hypothesis? What are the theoretical results pertaining to manifold hypothesis in the other papers we've been reading? Can we establish any analogs?]

[Emile says: What is the simplest possible way to empirically validate anything that we've constructed (after constructing it)?]

Theorem 7.2. *equivalency between effective dimension and energy?*

8 Examples

9 Conclusion

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